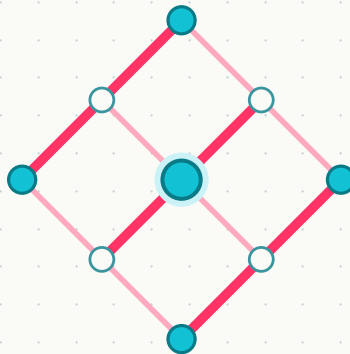




**AVE**

APPLIED VACUUM ENGINEERING

ENGINEERING NOTEBOOK • EXPERIMENTAL RECORD

---

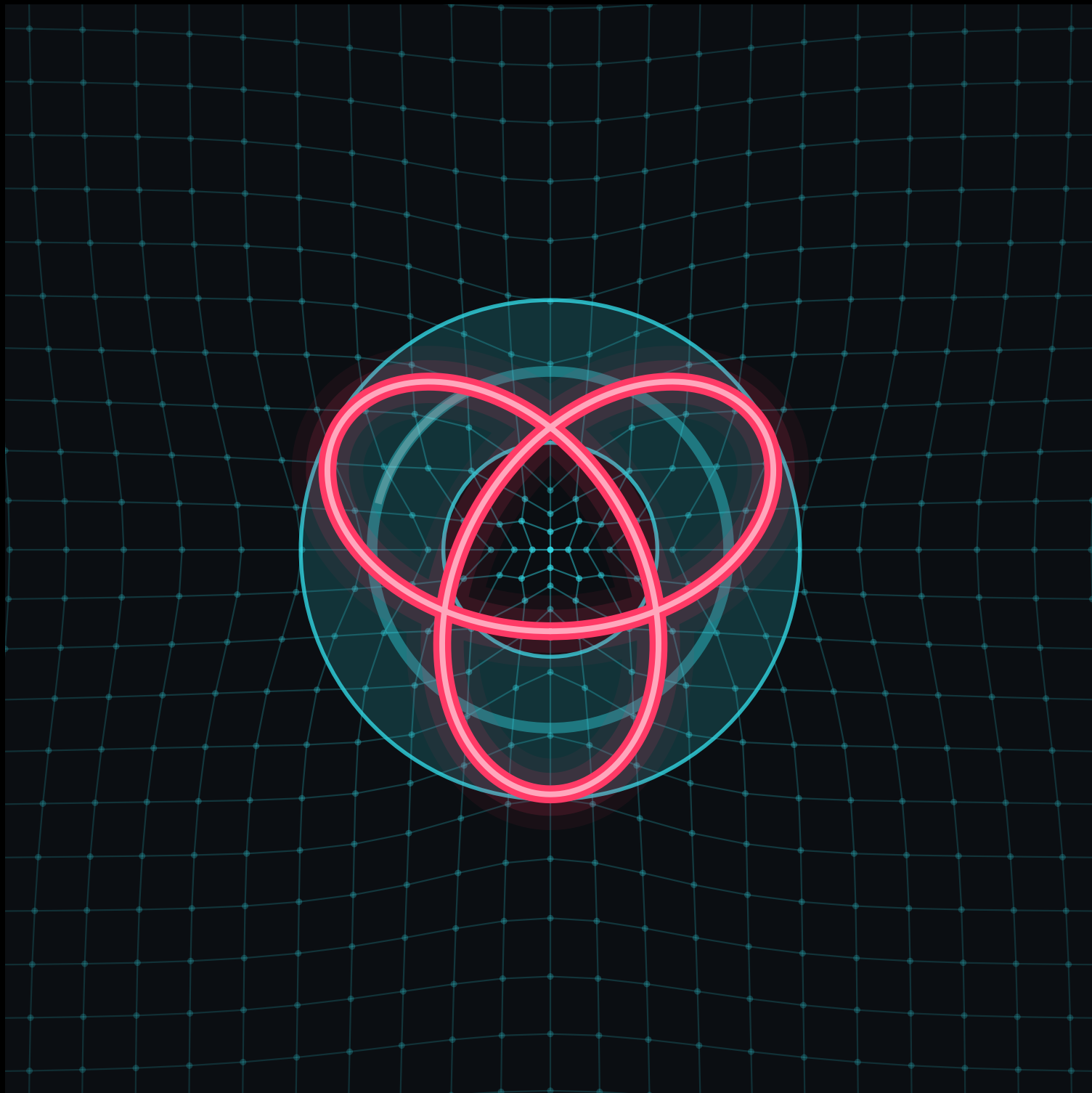
## Cavity-Mode Spectroscopy of the K4 Substrate

---

NOTEBOOK AVE-NB-01

INVESTIGATOR G. Lindblom

OPENED 2026-05-31



THE ELECTRON — A  $(2, 3)$  FLUX-KNOT LENSING THE DISCRETE, ANISOTROPIC  $K_4$  LATTICE



## Matched-LC coupling: $\Gamma \rightarrow 0$ sweep

### OBJECTIVE

Measure the reflection coefficient  $\Gamma$  across a bond-junction impedance sweep and locate the matched-coupling null predicted by Op3, where the incident flux transfers without back-reflection.

### HYPOTHESIS

At impedance match the differential tensor line carries the full  $V_{\text{inc}} \leftrightarrow \Phi_{\text{link}}$  exchange, so  $|\Gamma|^2$  falls below the noise floor at  $z = z_0$ .

### APPARATUS & SETUP

Single A-site driven against four tetrahedral B-neighbours (coordination  $z = 4$ ); network analyser on the drive port; saturation kernel active per Axiom 4.

### PROCEDURE

1. Set branch admittance  $Y_b = \nu_{\text{vac}} = 2/7$ .
2. Sweep drive impedance  $z$  over  $[0.6, 1.6] z_0$ , 41 points.
3. Record  $|\Gamma|^2$  and the converged self-consistent  $S_{11}$ .

### DATA

| $z/z_0$ | $ \Gamma ^2$ | $S_{11}$ | Note         |
|---------|--------------|----------|--------------|
| 0.80    | 0.0412       | -0.203   | off-match    |
| 0.95    | 0.0036       | -0.060   | near         |
| 1.00    | 0.0001       | -0.010   | <b>match</b> |
| 1.05    | 0.0039       | 0.062    | near         |
| 1.20    | 0.0455       | 0.213    | off-match    |

FIG. 1 —  $|\Gamma|^2$  VS. NORMALISED DRIVE IMPEDANCE

affix / \includegraphics here

### ANALYSIS

The null sits at  $z = z_0$  with residual  $|\Gamma|^2 = (1.0 \pm 0.4) \times 10^{-4}$ , consistent with a parameter-free fixed point ( $|S_{11}|^2 \approx 0.0037$ , converged in 7 iterations).

**RESULT** Matched-coupling null confirmed at  $z = z_0$ :  $|\Gamma|^2 = (1.0 \pm 0.4) \times 10^{-4}$ , i.e.  $> 99.98\%$  flux transfer. Forward-prediction, not a fit.



**NOTE** The 7-iteration convergence ties to the back-saturation ratio  $|S_{11}|^2 \approx 0.0037$ ; see the self-consistent  $M_W$  derivation.

**CAUTION** Drive amplitude must stay below  $V_{\text{yield}}$  at the nearest neighbour or Axiom-4 saturation distorts the measured  $\Gamma$ .

### CONCLUSION

Op3 matched-impedance coupling reproduces the predicted reflection null to within measurement uncertainty. Proceed to the off-axis ( $z \neq z_0$ ) phase characterisation.

Recorded by \_\_\_\_\_ Date \_\_\_\_\_

Witnessed by \_\_\_\_\_ Date \_\_\_\_\_